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United States Patent	12401260
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Hu; Yusheng et al.

Magnetism modulating ring structure, magnetic gear assembly

Abstract

Provided are a magnetism modulating ring structure, a magnetic gear assembly, and a compound motor. The magnetism modulating ring structure includes a plurality of modulating units and a plurality of connection parts. Each two adjacent modulating units are connected by one of the plurality of connection parts to form the magnetism modulating ring structure, and the magnetism modulating ring structure is arranged within an annular gap enclosed by a first rotor structure and a second rotor structure. A groove structure is formed on one side of the modulating unit facing the first rotor structure, and two sides of the groove structure respectively form a curved boot-like part, and an edge of the curved boot-like part facing the second rotor structure has a curved contour line parallel to flux lines passing through an inside of the curved boot-like part.

Inventors:	Hu; Yusheng (Zhuhai, CN), Chen; Bin (Zhuhai, CN), Ma; Xiaohao (Zhuhai, CN), Li; Quanfeng (Zhuhai, CN), Gui; Pengqian (Zhuhai, CN)
Applicant:	GREE ELECTRIC APPLIANCES, INC. OF ZHUHAI (Guangdong, CN)
Family ID:	1000008780498
Assignee:	GREE ELECTRIC APPLIANCES, INC. OF ZHUHAI (Guangdong, CN)
Appl. No.:	18/020941
Filed (or PCT Filed):	July 30, 2021
PCT No.:	PCT/CN2021/109495
PCT Pub. No.:	WO2022/110866
PCT Pub. Date:	June 02, 2022

Prior Publication Data

Document Identifier

US 20230283160 A1

Publication Date

Sep. 07, 2023

Foreign Application Priority Data

CN

202011376814.7

Nov. 30, 2020

Publication Classification

Int. Cl.: H02K49/10 (20060101)

U.S. Cl.:

CPC H02K49/102 (20130101);

Field of Classification Search

CPC: H02K (49/102); H02K (49/104); H02K (49/10); H02K (7/116); F16H (49/005)

USPC: 310/103; 310/104

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a U.S. National Stage of International Application No. PCT/CN2021/109495 filed on Jul. 30, 2021 entitled Magnetism Modulating Ring Structure Magnetic Gear Assembly, and Compound Motor and published as WO 2022/110866 on Jun. 2, 2022, which claims priority to Chinese Patent Application No. 202011376814.7, filed on Nov. 30, 2020, entitled “MAGNETISM MODULATING RING STRUCTURE, MAGNETIC GEAR ASSEMBLY, AND COMPOUND MOTOR”, the disclosures of which are incorporated herein by reference in their entireties. Every patent application and publication listed in this paragraph is hereby incorporated by reference in its entirety, as an example.

TECHNICAL FIELD

(2) The present disclosure relates to the field of contactless transmission equipment, and more particularly, to a magnetism modulating ring structure, a magnetic gear assembly, and a compound motor.

BACKGROUND

(3) The magnetic gear assembly generally includes an inner rotor, a magnetism modulating ring, and an outer rotor. Permanent magnets are provided on the outer peripheral surface of the inner rotor and the inner peripheral surface of the outer rotor, and the magnetism modulating ring is an annular structure formed by a plurality of pieces of iron cores assembled at equal intervals. However, the iron core in the conventional magnetism modulating ring has a rectangular-like structure, and the magnetism modulating ring provided with the rectangular-like structure has a limited effect on the magnetic field modulation, and the magnetic leakage between the iron cores is severe, thus reducing the output torque of the magnetic gear assembly. In addition, during manufacture of the magnetism modulating ring, a flux bridge needs to be added to serve as a connection between two adjacent rectangle-like iron cores, and the flux bridge causes relatively severe magnetic leakage due to an unreasonable structure of the flux bridge, thus causing the output torque of the magnetic gear assembly to be further reduced.

SUMMARY

(4) The main objective of the present disclosure is to provide a magnetism modulating ring structure, a magnetic gear assembly, and a compound motor to solve the problems in the related art that the iron core on the magnetism modulating ring has a rectangular-like structure, and that the magnetism modulating ring provided with the rectangular-like structure has a limited effect on the magnetic field modulation, and that the magnetic leakage between the iron cores is severe, thus reducing the output torque of the magnetic gear assembly.

(5) In order to achieve the above objective, according to one aspect of the present disclosure, a magnetism modulating ring structure is provided, and includes a plurality of modulating units and a plurality of connection parts, wherein each two adjacent modulating units are connected by one of the plurality of connection parts to form the magnetism modulating ring structure, and the magnetism modulating ring structure is arranged within an annular gap enclosed by a first rotor structure and a second rotor structure. A groove structure is formed on one side of the modulating unit facing the first rotor structure, and two sides of the groove structure respectively form a curved boot-like part, and an edge of the curved boot-like part facing the second rotor structure has a curved contour line parallel to flux lines passing through an inside of the curved boot-like part.

(6) In some embodiments, adjacent curved boot-like parts are connected by a connection part.

(7) In some embodiments, the plurality of connection parts are made of magnetic material. Two

adjacent curved boot-like parts and the connection part form a flux bridge structure. One end of the curved boot-like part away from the connection part forms a flux bridge head of the flux bridge structure, and the connection part forms a flux bridge center of the flux bridge structure.

(8) In some embodiments, t_1 denotes a thickness of the flux bridge head of the flux bridge structure, t_2 denotes a thickness between an outer circumferential surface of the modulating unit and a boot bottom of the curved boot-like part, and t_1 and t_2 satisfy: $0.25 \leq t_1/t_2 \leq 0.3$.

(9) In some embodiments, t_3 denotes a thickness of the flux bridge center of the flux bridge structure, and t_3 satisfies: $t_3 \leq 0.5$ mm.

(10) In some embodiments, a groove wall surface of the groove structure is a first curved face, and r_1 denotes a radius of curvature of the first curved face. A smooth transition is arranged between a side surface of the curved boot-like part facing the second rotor structure and a side surface of the connection part facing the second rotor structure, to form a curved transitional surface. A curved contour line of the curved transitional surface is parallel to the flux lines passing through the inside of the curved boot-like part. r_2 denotes a radius of curvature of the curved transitional surface, and r_1 and r_2 satisfy: $r_2 = 5 \times r_1$.

(11) In some embodiments, each modulating unit has a first side and a second side which are arranged to be opposite to each other; a_1 denotes an included angle formed between the first side and the second side of each modulating unit, and a_2 denotes an included angle formed between the first side of one modulating unit and the first side of the other modulating unit of two adjacent modulating units; a_1 and a_2 satisfy: $0.4 \leq a_1/a_2 \leq 0.5$; and r_1 and a_1 satisfy: $0.2 \leq r_1/a_1 \leq 0.3$.

(12) In some embodiments, the plurality of connection parts are made of nonmagnetic material; t_4 denotes a thickness of one end of the curved boot-like part away from the connection part; t_5 denotes a thickness between an outer circumferential surface of the modulating unit and a boot bottom of the curved boot-like part; and t_4 and t_5 satisfy: $0.25 \leq t_4/t_5 \leq 0.3$.

(13) In some embodiments, a boot face of the curved boot-like part is a second curved face, and r_3 denotes a radius of curvature of the second curved face; a groove wall surface of the groove structure is a third curved face, and r_4 denotes a radius of curvature of the third curved face; and r_3 and r_4 satisfy: $r_3 = 4 \times r_4$.

(14) In some embodiments, each modulating unit has a first side and a second side which are arranged to be opposite to each other; a_3 denotes an included angle formed between the first side and the second side of each modulating unit; a_4 denotes an included angle formed between boot heads of two curved boot-like parts of each modulating unit; a_3 and a_4 satisfy: $0.5 \leq a_3/a_4 \leq 0.6$; and r_4 and a_3 satisfy: $0.2 \leq r_4/a_3 \leq 0.3$.

(15) In some embodiments, the plurality of connection parts and the plurality of modulating units are formed integrally.

(16) According to another aspect of the present disclosure, a magnetic gear assembly is provided and includes the first rotor structure, the second rotor structure, and the magnetism modulating ring structure. The first rotor structure is sleeved on an outer peripheral side of a rotational shaft structure, and a first magnetic member is provided on an outer peripheral surface of the first rotor structure. The second rotor structure is sleeved over an outer peripheral side of the first rotor structure, and a second magnetic member is provided on an inner peripheral surface of the second rotor structure. The magnetism modulating ring structure is arranged in an annular gap enclosed by the first magnetic member and the second magnetic member, and the magnetism modulating ring structure is the magnetism modulating ring structure described above.

(17) In some embodiments, a rotational speed of the first rotor structure is greater than a rotational speed of the magnetism modulating ring structure, and a rotational speed of the second rotor structure is zero.

(18) In some embodiments, a gap is arranged between the magnetism modulating ring structure and the first magnetic member, and a gap is arranged between the magnetism modulating ring structure and the second magnetic member.

(19) In some embodiments, multiple first magnetic members are arranged, and the multiple first magnetic members are arranged at intervals along a circumferential direction of the first rotor structure. And/or multiple second magnetic members are provided, and the multiple second magnetic members are arranged at intervals along a circumferential direction of the second rotor structure.

(20) According to another aspect of the present disclosure, a compound motor is provided, and the compound motor includes a magnetic gear assembly, and the magnetic gear assembly is the magnetic gear assembly described above.

(21) In the technical solutions of the present invention, by modifying the structure of the modulating unit, each modulating unit has the groove structure and two curved boot-like parts. During an operation of the magnetic gear assembly, the groove structure can effectively reduce a variation amplitude of the magnetic density harmonic between the modulating units, thereby reducing the eddy current loss as much as possible. In addition, since an edge of the curved boot-like part has a curved contour line parallel to the flux lines passing through the inside of the curved boot-like part as possible, thus ensuring as many flux lines as possible to pass smoothly, thereby reducing the magnetic leakage between two adjacent modulating units effectively, and greatly improving the output torque of the magnetic gear assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings constituting a portion of the present disclosure are used to make the present disclosure to be further understood. The exemplary embodiments of the present disclosure and the description thereof are used to illustrate the present disclosure, but not intended to be construed as improper limitations on the present disclosure. In the accompanying drawings:

(2) FIG. 1 shows a schematic structural view of a magnetic gear assembly mounted on a rotational shaft structure according to an alternative embodiment of the present disclosure;

(3) FIG. 2 shows a schematic view of a magnetism modulating ring structure of the magnetic gear assembly in FIG. 1, in which a connection part and a modulating unit are integrally formed;

(4) FIG. 3 shows a schematic view of the magnetism modulating ring structure of the magnetic gear assembly in FIG. 1, in which the connection part made of nonmagnetic material is omitted;

(5) FIG. 4 shows a curve of an output torque of the magnetic gear assembly in FIG. 1 changing with t_1/t_2 ;

(6) FIG. 5 shows a curve of the output torque of the magnetic gear assembly in FIG. 1 changing with a_1/a_2 ;

(7) FIG. 6 shows a curve of an eddy current loss of the magnetic gear assembly in FIG. 1 changing with r_1/a_1 ;

(8) FIG. 7 shows a curve of the output torque of the magnetic gear assembly in FIG. 1 changing with r_1/a_1 ;

(9) FIG. 8 is a view showing a comparison between an output torque of a flux bridge structure of an alternative embodiment of the present disclosure and an output torque of an existing flux bridge structure;

(10) FIG. 9 shows flux lines of the magnetic gear assembly distributed in the magnetism modulating ring structure according to an alternative embodiment of the present disclosure;

(11) FIG. 10 shows a comparison of radial magnetic density harmonic distributions at an inner air gap;

(12) FIG. 11 shows a comparison of tangential magnetic density harmonic distributions at the inner air gap; and

(13) FIG. 12 is a view showing a comparison between an output torque of a magnetism modulating

ring structure of an alternative embodiment of the present disclosure and an output torque of an existing magnetism modulating ring structure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(14) The technical solutions in the embodiments of the present disclosure will be described clearly and completely hereinafter by combining with the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only part of embodiments rather than all embodiments of the present disclosure. The following description of at least one exemplary embodiment is merely illustrative but not intended to limit the present disclosure and any application or use of the present disclosure. Based on the embodiments in the present disclosure, all other embodiments obtained by those ordinary skilled in the art without involving any inventive efforts are within the scope of protection of the present disclosure.

(15) In order to solve the problems in the related art that the iron core on the magnetism modulating ring has a rectangular-like structure, and that the magnetism modulating ring provided with the rectangular-like structure has a limited effect on the magnetic field modulation, and that the magnetic leakage between the iron cores is severe, thus reducing the output torque of the magnetic gear assembly, the present disclosure provides a magnetism modulating ring structure, a magnetic gear assembly, and a compound motor. The compound motor includes the magnetic gear assembly, and the magnetic gear assembly is the magnetic gear assembly described herein.

(16) As shown in FIG. 1, the magnetic gear assembly includes a first rotor structure **10**, a second rotor structure **30**, and a magnetism modulating ring structure **40**. The first rotor structure **10** is sleeved on an outer peripheral side of the rotational shaft structure **20**, and a first magnetic member **11** is provided on an outer peripheral surface of the first rotor structure **10**. The second rotor structure **30** is sleeved over an outer peripheral side of the first rotor structure **10**, and a second magnetic member **31** is arranged on the inner peripheral surface of the second rotor structure **30**. The magnetism modulating ring structure **40** is arranged in an annular gap **100** enclosed by the first magnetic member **11** and the second magnetic member **31**, and the magnetism modulating ring structure **40** is the magnetism modulating ring structure described herein.

(17) It should be noted that in the embodiments of the present disclosure, a rotation speed of the first rotor structure **10** is greater than a rotation speed of the magnetism modulating ring structure **40**, and a rotation speed of the second rotor structure **30** is zero.

(18) It should be noted that in the embodiments of the present disclosure, a gap is arranged between the magnetism modulating ring structure **40** and the first magnetic member **11**, and a gap is arranged between the magnetism modulating ring structure **40** and the second magnetic member **31**, thereby enabling the magnetic gear assembly to realize a contactless torque transmission.

(19) As shown in FIG. 1, multiple first magnetic members **11** are provided, and the multiple first magnetic members **11** are arranged at intervals along a circumferential direction of the first rotor structure **10**, and/or multiple second magnetic members **31** are provided, and the multiple second magnetic members **31** are arranged at intervals along a circumferential direction of the second rotor structure **30**.

First Embodiment

(20) As shown in FIG. 2, the magnetism modulating ring structure includes a plurality of modulating units **41**, and each two adjacent modulating units **41** are connected by a connection part **42** to form the magnetism modulating ring structure, and the magnetism modulating ring structure is arranged within the annular gap **100** enclosed by the first rotor structure **10** and the second rotor structure **30**. A groove structure **411** is formed on one side of the modulating unit **41** facing the first rotor structure **10**, and two sides of the groove structure **411** respectively form a curved boot-like part **412**.

(21) By modifying the structure of the modulating unit **41**, each modulating unit **41** has the groove structure **411** and two curved boot-like parts **412**. During an operation of the magnetic gear assembly, the groove structure **411** can effectively reduce a variation amplitude of the magnetic

density harmonic between the modulating units **41**, thereby reducing the eddy current loss as much as possible. In addition, since an edge of the curved boot-like part **412** has a curved contour line **4121** parallel to the flux lines passing through the inside of the curved boot-like part **412** as possible, thus ensuring as many flux lines as possible to pass smoothly, thereby reducing the magnetic leakage between two adjacent modulating units **41** effectively, and greatly improving the output torque of the magnetic gear assembly.

(22) It should be noted that, in the present embodiment, in order to reduce the difficulty in manufacturing the magnetism modulating ring structure, alternatively, the adjacent curved boot-like parts **412** are connected by the connection parts. Further, the connection parts and the modulating units **41** are formed integrally, thereby ensuring subsequent overlaying and forming of the magnetism modulating ring structure.

(23) As shown in FIG. 2, the connection part is made of magnetic material, two adjacent curved boot-like parts **412** and the connection part form a flux bridge structure. One end of the curved boot-like part **412** away from the connection part forms a flux bridge head of the flux bridge structure, and the connection part forms a flux bridge center of the flux bridge structure. t_1 denotes a thickness of the flux bridge head of the flux bridge structure, and t_2 denotes a thickness between an outer circumferential surface of the modulating unit **41** and the boot bottom of the curved boot-like part **412**, and t_1 and t_2 satisfy: $0.25 \leq t_1/t_2 \leq 0.3$. In this case, by optimizing a ratio of the thickness t_1 of the flux bridge head to the thickness t_2 between the outer circumferential surface of the modulating unit **41** and the boot bottom of the curved boot-like part **412**, thereby avoiding the case that the flux lines cannot be guided better due to a too small ratio of the thickness t_1 of the flux bridge head to the thickness t_2 between the outer circumferential surface of the modulating unit **41** and the boot bottom of the curved boot-like part **412**, and avoiding a magnetic leakage phenomenon due to a too large ratio of the thickness t_1 of the flux bridge head to the thickness t_2 between the outer circumferential surface of the modulating unit **41** and the boot bottom of the curved boot-like part **412**.

(24) As shown in FIG. 2, t_3 denotes a thickness of the flux bridge center of the flux bridge structure, and t_3 satisfies: $T_3 \leq 0.5$ mm, thereby avoiding a severe magnetic leakage phenomenon due to a too large thickness of the flux bridge center of the flux bridge structure.

(25) As shown in FIG. 2, the groove wall surface of the groove structure **411** is a first curved face, and r_1 denotes a radius of curvature of the first curved face. A smooth transition is arranged between a side surface of the curved boot-like part **412** facing the second rotor structure **30** and a side surface of the connection part facing the second rotor structure **30**, to form a curved transitional surface. A curved contour line of the curved transitional surface is parallel to the flux lines passing through the inside of the curved boot-like part **412**. r_2 denotes a radius of curvature of the curved transitional surface, and r_1 and r_2 satisfy: $r_2 = 5 \times r_1$. In this case, flux lines can be ensured to pass through the flux bridge structure smoothly as many as possible, thereby reducing the magnetic leakage as much as possible.

(26) As shown in FIG. 2, each modulating unit **41** has a first side and a second side which are arranged to be opposite to each other. a_1 denotes an included angle formed between the first side and the second side of each modulating unit **41**, and a_2 denotes an included angle formed between the first side of one modulating unit **41** and the first side of the other modulating unit **41** of two adjacent modulating units **41**, and a_1 and a_2 satisfy: $0.4 \leq a_1/a_2 \leq 0.5$, and r_1 and a_1 satisfy: $0.2 \leq r_1/a_1 \leq 0.3$, thereby avoiding the case that the modulating units **41** are distributed to be too dense to guide the flux lines.

Second Embodiment

(27) It should be noted that, as shown in FIG. 3, the present embodiment differs from the first embodiment in that: the connection part is made of nonmagnetic material; t_4 denotes a thickness of one end of the curved boot-like part **412** away from the connection part, and t_5 denotes a thickness between the outer circumferential surface of the modulating unit **41** and the boot bottom of the

curved boot-like part **412**, and **t4** and **t5** satisfy: $0.25 \leq t4/t5 \leq 0.3$. In this case, the magnetic leakage caused by the presence of the flux bridge structure may be reduced as much as possible, thereby ensuring the output torque of the magnetic gear assembly to be as large as possible.

(28) It should be noted that in the present embodiment, in order to connect all modulating units **41** conveniently to form the magnetism modulating ring structure, all the modulating units **41** are connected by the connection parts made of nonmagnetic material, which facilitates not only the subsequent overlaying and forming but also an installation of the magnetism modulating ring structure.

(29) As shown in FIG. 3, a boot face of the curved boot-like part **412** is a second curved face, and **r3** denotes a radius of curvature of the second curved face. The groove wall surface of the groove structure **411** is a third curved face, and **r4** denotes a radius of curvature of the third curved face, and **r3** and **r4** satisfy: $r3=4 \times r4$. In this case, the curved boot-like part **412** is ensured to guide the flux lines as many as possible, thereby ensuring that the flux lines can pass through the curved boot-like part **412** smoothly.

(30) As shown in FIG. 3, each modulating unit **41** has a first side and a second side which are arranged to be opposite to each other. **a3** denotes an included angle formed between the first side and the second side of each modulating unit **41**, and **a4** denotes an included angle formed between boot heads **4122** of two curved boot-like parts **412** of each modulating unit **41**, and **a3** and **a4** satisfy: $0.5 \leq a3/a4 \leq 0.6$, and **r4** and **a3** satisfy: $0.2 \leq r4/a3 \leq 0.3$. In this case, flux lines are ensured to pass through the curved boot-like parts **412** smoothly, thereby reducing the magnetic leakage as much as possible, and further improving the output torque of the magnetic gear assembly.

(31) It should be noted that, in the embodiments of the present disclosure, each modulating unit **41** has the first side and the second side which are arranged to be opposite to each other. The first side and the second side may be named according to the counterclockwise direction of the magnetism modulating ring structure or may be named according to the clockwise direction of the magnetism modulating ring structure.

(32) FIG. 4 shows a curve of an output torque of the magnetic gear assembly changing with **t1/t2**. As can be seen from FIG. 4, when **t1/t2**=0.2, the output torque of the magnetic gear assembly reaches a peak of about 51 Nm. When **t1/t2**<0.2, the output torque of the magnetic gear assembly increases as the ratio **t1/t2** increases. When **t1/t2**>0.2, the output torque of the magnetic gear assembly decreases as the ratio **t1/t2** increases.

(33) FIG. 5 shows a curve of the output torque of the magnetic gear assembly changing with **a1/a2**. As can be seen from FIG. 5, when **a1/a2**<0.4, the output torque of the magnetic gear assembly increases as the ratio **a1/a2** increases. When $0.4 \leq a1/a2 \leq 0.6$, the curve of the output torque of the magnetic gear assembly changing with **a1/a2** is substantially flat. When **a1/a2**>0.6, the output torque of the magnetic gear assembly decreases as the ratio **a1/a2** increases.

(34) FIG. 6 shows a curve of an eddy current loss of the magnetic gear assembly changing with **r1/a1**. As can be seen from FIG. 6, when **r1/a1**=6, the eddy current loss of the magnetic gear assembly is the lowest. When **r1/a1**<6, the eddy current loss of the magnetic gear assembly decreases as the ratio **r1/a1** increases. When **r1/a1**>6, the eddy current loss of the magnetic gear assembly tends to increase as the ratio **r1/a1** increases.

(35) FIG. 7 shows a curve of the output torque of the magnetic gear assembly changing with **r1/a1**. As can be seen from FIG. 7, when **r1/a1**≤0.3, the output torque of the magnetic gear assembly tends to decrease smoothly as the ratio **r1/a1** increases. When **r1/a1**>0.3, the output torque of the magnetic gear assembly tends to decrease rapidly as the ratio **r1/a1** increases, and a slope of the latter is greater than a slope of the former.

(36) FIG. 8 is a view showing a comparison between the output torque of a flux bridge structure of the first embodiment and the output torque of an existing flux bridge structure. As can be seen from FIG. 8, the flux bridge connection manner provided by the present disclosure can ensure that the magnetic gear assembly outputs a larger output torque than the output torque of the magnetic gear

assembly of the existing flux bridge connection manner.

(37) FIG. 9 shows flux lines of the magnetic gear assembly distributed in the magnetism modulating ring structure of the first embodiment. In FIG. 9, A denotes flux lines.

(38) FIG. 10 shows a comparison of radial magnetic density harmonic distributions at an inner air gap.

(39) FIG. 11 shows a comparison of tangential magnetic density harmonic distributions at the inner air gap.

(40) FIG. 12 is a view showing a comparison between an output torque of a magnetism modulating ring structure of the first embodiment and an output torque of an existing magnetism modulating ring structure. The magnetism modulating ring structure provided by the present disclosure can ensure that the magnetic gear assembly outputs a larger output torque than the output torque of the magnetic gear assembly of the existing magnetism modulating ring structure.

(41) It should be noted that, the terminology herein is used for describing the specific embodiments, but not intended to limit the illustrative embodiments of the present disclosure. The singular terms used herein are intended to include their plural unless specific descriptions are provided in context. It should be also understood that, the terms “include” and/or “comprise” in the description refer to including the features, steps, operations, devices, components, and/or combinations thereof.

(42) Unless otherwise specified, the relative arrangement, numerical expressions and values of the components and steps described in these embodiments do not limit the scope of this disclosure. Moreover, it should be understood that, for convenience of description, the dimensions of the parts shown in the accompanying drawings are not in accordance with actual proportionality. The technologies, methods and devices known to those of ordinary skill in the relevant fields may not be discussed in detail, but, where appropriate, the technologies, the methods and the devices shall be considered as part of the allowed specification. In all the examples shown and discussed herein, any specific value should be interpreted as merely an example, not as a limitation. Other examples of exemplary embodiments may therefore have different values. It should be noted that similar reference numerals and letters denote similar terms in the following figures, so that once a particular term is defined in one of the figures, further discussion is not required in the subsequent drawings.

(43) For the convenience of description, terms of spatial relations such as “above”, “over”, “on a top surface”, “upper”, etc., may be used herein to describe the spatial position relationships of a device or a feature with other devices or features shown in the drawings. It should be understood that the terms of spatial relations are intended to include other different orientations in use or operation in addition to the orientation of the device described in the drawings. For example, if the device in the drawings is placed upside down, the device described as “above other devices or structures” or “over other devices or structures” will be positioned as “below other devices or structures” or “under other devices or structures”. Thus, the exemplary term “above” may include both “above” and “below”. The device can also be positioned in other different ways (rotating 90 degrees or at other orientations), and the corresponding explanations for the description of the spatial relations will be provided herein.

(44) It should be noted that, the terminology herein is used for describing the specific embodiments, but not intended to limit the illustrative embodiments of the present disclosure. The singular terms used herein are intended to include their plural unless specific descriptions are provided in context. It should be also understood that, the terms “include” and/or “comprise” in the description refer to including the features, steps, operations, devices, components, and/or combinations thereof.

(45) It should be specified that the terms “first”, “second”, etc. in the description, the claims and the drawings in the present disclosure are just used to distinguish similar objects, but not used to describe a specific order or an order of priority. It should be understood that such terms may be

interchangeable under appropriate conditions, such that the embodiments of the present disclosure illustrated in the drawing or described herein can be implemented, for example, in a sequence other than the sequences illustrated or described herein.

(46) What described above are some embodiments of the present disclosure, but not intended to limit the present disclosure. For those skilled in the art, various amendments and modifications can be made. Any modifications, equivalent substitutions and improvements made within the spirits and principles of the present disclosure are all within the scope of protection of the present disclosure.

Claims

1. A magnetism modulating ring structure, comprising: a plurality of modulating units and a plurality of connection parts; wherein each two adjacent modulating units are connected by one of the plurality of connection parts to form the magnetism modulating ring structure, and the magnetism modulating ring structure is arranged within an annular gap enclosed by a first rotor structure and a second rotor structure; a groove structure is formed on one side of the modulating unit facing the first rotor structure, and two sides of the groove structure respectively form a curved boot-like part, and an edge of the curved boot-like part facing the second rotor structure has a curved contour line parallel to flux lines passing through an inside of the curved boot-like part; each modulating unit has a first side and a second side which are arranged to be opposite to each other, one curved boot-like part of the first side of one modulating unit of two adjacent modulating units and another curved boot-like part of the second side of the other modulating unit of the two adjacent modulating units are connected to a same connection part; and the plurality of connection parts are made of magnetic material; two adjacent curved boot-like parts connected to the same connection part form a flux bridge structure; one end of the curved boot-like part away from the connection part forms a flux bridge head of the flux bridge structure, and the connection part forms a flux bridge center of the flux bridge structure.
2. The magnetism modulating ring structure according to claim 1, wherein $t1$ denotes a thickness of the flux bridge head of the flux bridge structure, $t2$ denotes a thickness between an outer circumferential surface of the modulating unit and a boot bottom of the curved boot-like part, and $t1$ and $t2$ satisfy: $0.25 \leq t1/t2 \leq 0.3$.
3. The magnetism modulating ring structure according to claim 1, wherein $t3$ denotes a thickness of the flux bridge center of the flux bridge structure, and $t3$ satisfies: $t3 \leq 0.5$ mm.
4. The magnetism modulating ring structure according to claim 1, wherein a groove wall surface of the groove structure is a first curved face, and $r1$ denotes a radius of curvature of the first curved face; a smooth transition is arranged between a side surface of the curved boot-like part facing the second rotor structure and a side surface of the connection part facing the second rotor structure, to form a curved transitional surface; a curved contour line of the curved transitional surface is parallel to the flux lines passing through the inside of the curved boot-like part; and $r2$ denotes a radius of curvature of the curved transitional surface, and $r1$ and $r2$ satisfy: $r2 = 5 \times r1$.
5. The magnetism modulating ring structure according to claim 4, wherein $a1$ denotes an included angle formed between the first side and the second side of each modulating unit, and $a2$ denotes an included angle formed between the first side of one modulating unit and the first side of the other modulating unit of two adjacent modulating units; and $a1$ and $a2$ satisfy: $0.4 \leq a1/a2 \leq 0.5$.
6. The magnetism modulating ring structure according to claim 1, wherein the plurality of connection parts and the plurality of modulating units are formed integrally.
7. A magnetic gear assembly, comprising: the first rotor structure, wherein the first rotor structure is configured to be sleeved on an outer peripheral side of a rotational shaft structure, and a first magnetic member is provided on an outer peripheral surface of the first rotor structure; the second rotor structure, wherein the second rotor structure is sleeved over an outer peripheral side of the

first rotor structure, and a second magnetic member is provided on an inner peripheral surface of the second rotor structure; and the magnetism modulating ring structure of claim 1, wherein the magnetism modulating ring structure is arranged in an annular gap enclosed by the first magnetic member and the second magnetic member.

8. The magnetic gear assembly according to claim 7, wherein a rotational speed of the first rotor structure is greater than a rotational speed of the magnetism modulating ring structure, and a rotational speed of the second rotor structure is zero.

9. The magnetic gear assembly according to claim 8, wherein a gap is arranged between the magnetism modulating ring structure and the first magnetic member, and a gap is arranged between the magnetism modulating ring structure and the second magnetic member.

10. The magnetic gear assembly according to claim 7, wherein a gap is arranged between the magnetism modulating ring structure and the first magnetic member, and a gap is arranged between the magnetism modulating ring structure and the second magnetic member.

11. The magnetic gear assembly according to claim 7, wherein multiple first magnetic members are arranged, and the multiple first magnetic members are arranged at intervals along a circumferential direction of the first rotor structure.

12. The magnetic gear assembly according to claim 11, wherein multiple second magnetic members are provided, and the multiple second magnetic members are arranged at intervals along a circumferential direction of the second rotor structure.

13. The magnetic gear assembly according to claim 7, wherein multiple second magnetic members are provided, and the multiple second magnetic members are arranged at intervals along a circumferential direction of the second rotor structure.
